SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks **COURSE CODE: CSA0716**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

ANSWERS

Case Study 1

Q1. Analyse whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.
Answer:
The main cause is the transport protocol (UDP) along with the application layer buffering strategy. Real-time video conferencing generally uses UDP because it provides fast transmission without waiting for lost packets. However, the reported 8% packet loss, 180 ms jitter, and fluctuating bandwidth cause missing audio/video packets. If the application buffer is too small, it cannot compensate for jitter, resulting in frozen video frames and distorted voice.
Analysis:

- Packet Loss = 8% (Very High)
- Jitter = 180 ms (Very Poor)
- Recommended Jitter < 30 ms
- Therefore, the issue is caused by both UDP packet loss and poor application buffering.

Q2. Justify your answer based on the communication requirements of real-time multimedia applications.
Answer:
Real-time applications require:

- Very low latency (<150 ms)
- Low jitter (<30 ms)
- Continuous packet delivery
- Minimal retransmissions

TCP performs retransmissions, increasing delay, making it unsuitable for live conferencing. UDP delivers packets immediately without retransmission, ensuring lower latency. However, excessive packet loss affects media quality. Therefore, applications use UDP with adaptive buffering and error correction instead of TCP.

Q3. Recommend suitable protocol-level or application-level improvements.
Answer:
Protocol-Level Improvements

1. Use RTP over UDP.
2. Enable Forward Error Correction (FEC).
3. Apply QoS (Quality of Service).

Application-Level Improvements

1. Adaptive jitter buffer.
2. Adaptive bitrate streaming.
3. Dynamic codec selection.
4. Congestion monitoring.

Expected Result

- Packet Loss: 8% → <1%
- Jitter: 180 ms → <30 ms
- Smooth video and clear voice.

Case Study 2

Q1. Analyze the reasons behind the high DNS resolution time.
Answer:
DNS lookup time is 800 ms, which is much higher than the acceptable value (<50 ms).
Reasons:

- Centralized DNS server.
- High server workload.
- Long network distance.
- Inefficient DNS caching.
- Improper TTL configuration.

Calculation
Current DNS Time = 800 ms
Target DNS Time = 50 ms
Reduction Required
= 800 − 50
= 750 ms
Percentage Improvement
= (750 / 800) × 100
= 93.75%

Q2. Propose a suitable DNS architecture.
Answer:
Design:

- Deploy Anycast DNS servers globally.
- Enable DNS pre-fetching.
- Optimize TTL values (300–600 seconds).
- Use recursive DNS caching.
- Load balancing among DNS servers.

Expected lookup time:
Current = 800 ms
Estimated after optimization:

- Anycast = 25 ms
- Cache = 10 ms
- Network delay = 10 ms

Total ≈ 45 ms
Target achieved (<50 ms).

Q3. Evaluate advantages and limitations.
Advantages
- Faster DNS resolution.
- High availability.
- Better scalability.
- Automatic failover.
- Lower latency.

Limitations
- Higher deployment cost.
- Complex configuration.
- Cache synchronization challenges.

Overall, performance and availability improve significantly.

Case Study 3

Q4. Examine the impact of centralized DNS architecture.
Answer:
A centralized DNS server causes:
- High response time.
- Single point of failure.
- Increased server overload.
- Poor user experience.
- Reduced application availability.

Calculation
Current DNS Delay = **800 ms**
Desired Delay = **50 ms**
Latency Reduction
= 800 − 50
= **750 ms**
Improvement
= **93.75%**

Q5. Design a distributed DNS solution.
Answer:
Recommended Architecture:
- Global Anycast DNS servers.
- Regional DNS replicas.
- Recursive DNS caching.
- Load balancing.
- Automatic failover.

Benefits:
- Faster response.
- Global scalability.
- Fault tolerance.
- Better reliability
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Q6. Analyze TTL, Anycast routing and DNS caching.
Answer:
TTL
- Stores DNS records locally.
- Reduces repeated lookups.

Anycast
- Sends users to the nearest DNS server.
- Minimizes latency.

DNS Cache
- Stores previous queries.
- Reduces server workload.

Together these techniques reduce lookup time, improve fault tolerance, and increase availability.

Case Study 4

Q1. Analyze three TCP-related factors responsible for latency increase.
Answer:
1. Flow Control
    - Small receive window limits data transmission.
    - Sender waits for acknowledgements.
2. Nagle's Algorithm
    - Combines small packets before transmission.
    - Introduces delay in stock trading.
3. SYN Queue Limitation
    - Too many connection requests overflow the queue.
    - Causes delayed connection establishment.

Q2. Explain how each factor affects transaction processing.
Answer:
Flow Control
- Slows order processing.

Nagle's Algorithm
- Delays small order messages.

SYN Queue Overflow
- Increases connection setup delay.
- Customers receive confirmations late.

Calculation
Current latency = 40 ms
Required latency = 1 ms
Reduction
= 40 − 1
= 39 ms
Percentage Improvement
= (39 / 40) × 100
= 97.5%

Q3. Recommend TCP configuration changes.
Answer:
Recommendations:
1. Disable Nagle's Algorithm (TCP_NODELAY).
2. Increase TCP receive window size.
3. Increase SYN backlog queue.
4. Enable TCP Fast Open.
5. Optimize retransmission timeout.
6. Reduce unnecessary buffering.

Expected Result
- Latency reduced from **40 ms to approximately 1–5 ms**.
- Faster trade confirmations.
- Improved reliability and customer satisfaction.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understandi ng; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand